



FIELD INTELLIGENCE
TERRASATCH
LISTEN | WATCH | LEARN | ADAPT

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TerraSatch Inc.

Field intelligence for backcountry teams and remote operations

<https://www.terasatch.com/>



Executive Summary

TerraSatch is a field-intelligence platform for teams that operate where communications, terrain, weather, and human judgment matter more than perfect connectivity.

The platform is being built around a simple operating loop:

LISTEN. WATCH. LEARN. ADAPT.

LISTEN begins with the information field teams already produce: authorized radio traffic, spoken updates, observations, notes, location context, and other approved operational inputs. TerraListen is the current front door to the platform. It turns authorized radio and field input into searchable, source-linked records without requiring teams to abandon the radio workflows they already trust.

WATCH connects those records to place. A radio call or field observation becomes more useful when it is viewed alongside terrain, weather, forecast zones, operational boundaries, incident context, drone-derived cells, and other approved map layers. TerraSatch is designed to make the relationship between a field report and its surrounding operating picture visible and reviewable.

LEARN preserves what the field already learned. Calls, observations, terrain cells, documents, snowpits, incident records, shift notes, and review decisions can become durable operational memory instead of disappearing into disconnected radio traffic, notebooks, files, and one-off reports. The original observation remains distinct from AI interpretation.

ADAPT turns reviewed context into useful operational outputs. TerraSatch can prepare briefings, shift handoffs, alerts, reports, follow-up queues, map updates, and integration-ready records for approved human action. Qualified operators remain responsible for what is published, communicated, or acted on.

TerraSatch is not intended to replace forecasters, patrollers, firefighters, dispatchers, incident commanders, guides, engineers, SAR leaders, or other qualified professionals. It is decision-support and workflow infrastructure designed to reduce fragmentation, preserve context, and make field information easier to review.

The near-term commercial focus is **TerraListen**. The broader TerraSatch platform extends that radio-to-record workflow into terrain-aware operational intelligence through TerraSatch Edge, the TerraSatch API and control plane, Satchy, TerraGrid, and domain-specific modules including AvyTS, PyroTS, HydroTS, GeoTS, and InfraTS.

1. Why TerraSatch Exists

1.1 The field-information gap

Remote operations rarely suffer from a total lack of information. More often, the problem is that useful information is scattered across too many systems and disappears between handoffs.



A single operating period may involve:

- radio traffic;
- handwritten or digital field notes;
- dispatch records;
- weather and forecast products;
- terrain maps;
- GIS layers;
- drone imagery;
- photographs and measurements;
- team locations;
- incident timelines;
- shift-change conversations;
- PDF reports;
- spreadsheets;
- archived observations; and
- organization-specific knowledge held by experienced operators.

The individual pieces may be useful, but the workflow connecting them is often manual.

A field update can be heard over the radio, remembered by one person, written down somewhere else, mapped later, summarized at shift change, and finally copied again into a report. Every handoff creates another opportunity for information to be delayed, detached from its source, duplicated, or lost.

TerraSatch is designed to improve that middle layer.

1.2 The operating problem

The problem is not simply “more data.”

The operational question is:

How can a team preserve a trustworthy field signal, connect it to the right place and context, retain what was learned, and prepare a useful output without removing the human responsible for the decision?

TerraSatch answers that question through a repeatable operating loop:

LISTEN → WATCH → LEARN → ADAPT

2. The TerraSatch Operating Loop

2.1 LISTEN — Capture the field signal



TerraListen

TerraListen is TerraSatch's current AI-assisted radio and field-input product.

It is designed to work with authorized communications and approved operational data. The goal is not to change how crews communicate. The goal is to preserve useful information that would otherwise be trapped inside a transient voice channel or scattered across manual notes.

A typical TerraListen workflow is:

Radio or approved field input → source record → transcript → structured observation → location/context → timeline → summary/report-ready record → human review

Depending on the deployment, an event may include:

- organization and site;
- radio or input source;
- channel or configured listening profile;
- timestamp;
- original audio or approved source record;
- transcript;
- extracted entities and field terms;
- location or terrain reference;
- operational category;
- confidence and review state;
- related map cell or incident;
- linked documents or observations; and
- approved downstream actions.

Preserve the original

A core design rule is that the original field observation and the AI-assisted interpretation are not the same thing.

TerraSatch should preserve:

1. what was actually received or entered;
2. what the system inferred from it;
3. what a human reviewer confirmed, corrected, or rejected; and
4. what was ultimately published or acted on.

That separation supports auditability and reduces the risk of an AI-generated interpretation being mistaken for the original report.

TerraSatch Edge

TerraSatch Edge is the field-side component used to connect approved devices and listening workflows to the TerraSatch control plane.



Its role can include:

- pairing a field computer or edge device;
- associating the device with an approved organization and site;
- loading authorized configuration;
- reporting device and service health;
- managing approved listening or ingest profiles;
- forwarding selected events to the TerraSatch API; and
- supporting local processing where connectivity or deployment requirements call for it.

Edge deployments should be scoped to the organization's authorized frequencies, devices, retention requirements, and operating policies.

Noise, silence, and selective logging

Long-running radio environments create a practical problem: most receiver time is not meaningful speech.

TerraListen should not treat every second of receiver output as an operational event.

The intended workflow is to use configured signal detection, squelch, event segmentation, channel/source metadata, and additional filtering so that silence, empty receiver noise, and irrelevant artifacts are not blindly uploaded as records.

This supports lower processing cost, cleaner event history, and more useful review queues.

2.2 WATCH — Put the signal in context

A transcript by itself is useful, but a field report becomes more operationally valuable when it is tied to place.

WATCH is TerraSatch's mapping and situational-awareness layer.

It is designed to connect field records to the context a reviewer needs to understand what the report means.

That context may include:

- latitude and longitude;
- named terrain features;
- route segments;
- forecast or administrative zones;
- elevation;
- slope and aspect;
- weather;
- wind direction and loading context;
- recent observations;



- incidents;
- operational boundaries;
- team or asset locations;
- drone-derived terrain cells;
- wildfire perimeters;
- hydrologic context;
- infrastructure corridors; and
- organization-approved GIS layers.

Terrain cells and inspectable geography

TerraSatch uses terrain cells and other inspectable map geometries to organize local evidence without pretending that an entire mountain, drainage, fire area, or corridor behaves uniformly.

A cell can act as a durable container for:

- terrain characteristics;
- field observations;
- radio-linked events;
- drone or remote-sensing context;
- weather and forecast context;
- operational notes; and
- review status.

The intent is not to turn a cell into an autonomous hazard decision. It is to create a consistent place where evidence can accumulate and be inspected.

AvyTS

AvyTS applies the WATCH model to avalanche and snowpack operations.

The module is intended to connect:

- avalanche observations;
- radio reports;
- terrain cells;
- snowpit information;
- slope, aspect, and elevation;
- forecast-zone context;
- weather and wind;
- route and operational context; and
- reviewer decisions.

AvyTS should be described as an operational intelligence and decision-support layer, not an avalanche forecast authority and not a replacement for a qualified forecaster.

PyroTS



PyroTS applies the same field-intelligence loop to wildfire and wildland operations.

Potential inputs include:

- authorized radio events;
- incident and perimeter context;
- terrain;
- wind and weather;
- fuels and vegetation context;
- burn history;
- crew or resource updates;
- UAS or drone-derived cell information; and
- field observations.

The intended output is a clearer, reviewable operating picture. PyroTS is not a replacement for incident command, official fire-behavior products, or qualified fire personnel.

HydroTS, GeoTS, and InfraTS

The broader platform architecture supports additional terrain and infrastructure domains:

HydroTS — watershed, runoff, flooding, snowmelt, and water-related field context.

GeoTS — geologic terrain, rockfall, slope-change, and inspection-oriented context.

InfraTS — infrastructure, utilities, transportation, remote assets, inspection workflows, and hazard exposure.

These modules represent the broader platform direction. Their deployment depth should be described according to the specific pilot, integration, and validation status of each use case rather than as universally production-ready products.

WATCH is map-agnostic by design

TerraSatch should not require an organization to abandon its existing mapping or GIS system.

The platform is intended to:

- work with existing basemaps and provider geometries;
- retain organization-approved boundaries;
- export useful map and event data;
- expose integration-ready records through APIs; and
- act as an operational intelligence layer around tools teams already use.

2.3 LEARN — Build durable operational memory



Field teams repeatedly solve problems that their own systems fail to remember.

A useful radio report may exist for seconds. A shift handoff may exist for minutes. A PDF may be generated and then buried in a folder. A snowpit, image, incident note, or mapping change may be difficult to reconnect to the event that caused it.

LEARN is the part of TerraSatch that preserves those relationships.

Operational memory

TerraSatch is designed to connect:

- radio calls;
- transcripts;
- observations;
- map cells;
- terrain context;
- snowpits;
- weather context;
- drone-derived evidence;
- documents;
- incident timelines;
- review decisions;
- shift handoffs; and
- generated reports.

The objective is a searchable operational history that can be reviewed across:

- a single event;
- a shift;
- a day;
- a storm cycle;
- a fire;
- a season;
- a site; or
- a recurring operational workflow.

Satchy

Satchy is TerraSatch's AI agent and interface for working with approved field intelligence.

Satchy can assist with tasks such as:

- structuring an observation from a transcript;
- locating related evidence;
- summarizing what changed;
- preparing a briefing;
- identifying records that require review;



- organizing handoff information;
- finding related events or cells; and
- helping a user navigate the operational record.

Satchy should remain constrained by the same rules as the rest of the platform.

It should not silently rewrite source evidence, make autonomous go/no-go decisions, or hide the basis for an operational output.

Human review as part of learning

Human correction is not an exception to the system. It is part of the system.

A reviewer may:

- correct a transcript;
- refine a location;
- change an observation category;
- reject an AI extraction;
- attach additional evidence;
- add uncertainty;
- approve a record;
- update an event; or
- publish a reviewed output.

Those changes should become part of the record so future users can understand not only the final answer, but how the record evolved.

2.4 ADAPT — Prepare reviewed operational outputs

ADAPT is where preserved field context becomes something a team can use.

Potential outputs include:

- shift handoffs;
- incident timelines;
- daily or operational briefings;
- observation summaries;
- follow-up queues;
- map updates;
- reviewed alerts;
- PDF reports;
- structured exports;
- API records;
- GIS-ready data;
- archived transcripts; and



- organization-specific documents.

Review before action

TerraSatch is designed around reviewable outputs rather than autonomous operational control.

The platform may prepare an action, but the organization determines who is authorized to approve, publish, dispatch, or act.

A typical pattern is:

Field signal → structured record → contextual review → prepared output → authorized human action

Existing workflows first

ADAPT is also the integration principle of TerraSatch.

Organizations should not have to rebuild their entire operating environment around TerraSatch.

The platform is intended to support outputs that can fit into:

- existing GIS systems;
 - incident-management processes;
 - document workflows;
 - shift-change procedures;
 - reporting systems;
 - internal dashboards; and
 - partner APIs.
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3. Product Architecture

3.1 Field layer

The field layer is where information originates.

Examples include:

- handheld and mobile radio systems;
- approved SDR receivers;
- field computers;
- operator-entered observations;
- existing incident systems;
- weather and sensor feeds;



- approved documents;
- drone-derived data; and
- GIS layers.

TerraSatch does not assume ownership of every source. It is designed to connect approved sources into a more coherent operational record.

3.2 TerraSatch Edge

TerraSatch Edge provides deployment-side connectivity for approved field devices.

It can support:

- device registration;
- organization/site pairing;
- configuration retrieval;
- heartbeat and hardware state;
- audio or event ingest;
- local preprocessing;
- constrained-connectivity workflows; and
- controlled synchronization with the API.

3.3 TerraSatch API and control plane

The TerraSatch API is the shared control plane connecting:

- organizations;
- sites;
- devices;
- TerraSatch Edge;
- TerraListen ingest;
- event records;
- map-linked observations;
- operational timelines;
- reporting workflows; and
- approved integrations.

The API strategy is integration-first.

Public and partner-facing documentation should focus on the interfaces that are actually supported, including scoped credentials, organization/site/device configuration, ingest, event retrieval, and approved outputs.

The current public product direction emphasizes a documented API and OpenAPI-based reference rather than requiring organizations to adopt a proprietary client application.



3.4 Event and observation record

A TerraSatch operational record should be able to answer:

- What happened?
- When did it happen?
- Where did it happen?
- What source produced the information?
- What did the original source say?
- What did the system infer?
- What evidence is linked?
- What organization and site own the record?
- What review state is it in?
- Who changed it?
- What was published or generated from it?

This record is the bridge between LISTEN, WATCH, LEARN, and ADAPT.

3.5 Map and terrain layer

The map layer can combine:

- public basemaps;
- terrain derivatives;
- organization-approved boundaries;
- provider geometries;
- cells;
- observations;
- events;
- weather;
- operational zones;
- hazard context;
- drone-derived products; and
- other approved sources.

The map should make evidence easier to inspect, not hide it behind a single opaque score.

3.6 Reporting and document layer

Reports should be reproducible from reviewed records.

A generated output should retain enough provenance to identify:

- included observations;
- time window;
- reviewer state;



- map or cell context;
- source records;
- generation time; and
- organization/site ownership.

The goal is a defensible record that can be revisited after the shift or incident.

4. Current Product Stack

4.1 TerraListen — current product focus

Status: current commercial and pilot focus

TerraListen is the current entry point for organizations evaluating TerraSatch.

It focuses on converting approved radio or field inputs into a structured, reviewable operational record.

The simplest adoption path is intentionally narrow:

1. select one recurring workflow;
2. use approved sample data;
3. prove that the resulting record is useful;
4. measure the real processing and retention requirements; and
5. expand only when the workflow justifies the additional scope.

4.2 TerraSatch Edge

Status: active infrastructure component

TerraSatch Edge connects approved field hardware and local environments to TerraSatch services.

It is especially relevant where teams need:

- radio ingest;
- edge-side configuration;
- device health;
- site-specific profiles; or
- constrained-connectivity operation.

4.3 TerraSatch API

Status: active platform infrastructure



The API acts as the control plane for organizations, sites, devices, ingest, events, and reviewable records.

It is also the primary path for connecting TerraSatch to existing software instead of forcing replacement of established systems.

4.4 Satchy

Status: current TerraSatch AI agent identity

Satchy is the TerraSatch operational AI agent.

Historical TerraSatch materials may refer to **SherpAI**. Current public-facing material should use **Satchy** unless discussing historical product evolution.

4.5 Wasatch Relay

Status: earlier TerraListen MVP identity

Wasatch Relay was an earlier MVP expression of the radio-to-record workflow.

Current materials should describe the product as **TerraListen**. Wasatch Relay may be referenced when discussing the project's development history, but it should not compete with the current product name.

4.6 TerraGrid and terrain modules

Status: broader platform expansion

TerraGrid and the domain modules extend TerraListen's field-signal workflow into richer terrain and operational intelligence.

The current module family is:

- **AvyTS** — avalanche and snowpack;
- **PyroTS** — wildfire and wildland operations;
- **HydroTS** — hydrology and watershed context;
- **GeoTS** — geology, rockfall, and slope context; and
- **InfraTS** — infrastructure and remote asset operations.

The whitepaper should distinguish between the platform architecture, controlled demonstrations, active infrastructure, and future production scope.

5. Operational Use Cases



5.1 Avalanche forecasting and mountain operations

Potential workflows include:

- radio-based avalanche observations;
- wind-loading reports;
- cracking or collapsing reports;
- avalanche sightings;
- snowpit and snowpack observations;
- route conditions;
- visibility changes;
- control-work notes;
- patrol coordination;
- forecast-zone context;
- terrain-cell review;
- shift handoffs; and
- incident documentation.

TerraSatch should preserve published avalanche-center products as authoritative inputs where applicable. It should not present its own AI interpretation as a replacement for an official advisory or qualified forecaster.

5.2 Ski patrol and resort operations

Potential workflows include:

- patrol radio events;
- incidents;
- trail or terrain status;
- snow-safety observations;
- lift or infrastructure field updates;
- guest-related operational events;
- search activity;
- shift handoffs; and
- daily reports.

5.3 Search and rescue

Potential workflows include:

- team location updates;
- search-area references;
- assignment changes;
- subject or clue reports;
- medical or extraction updates;
- incident timelines;



- map-linked radio events; and
- handoff documentation.

5.4 Wildfire and wildland operations

Potential workflows include:

- radio-linked field observations;
- weather and wind context;
- mapped crew or resource updates;
- incident perimeter context;
- local terrain cells;
- UAS observations;
- safety-zone or route context;
- operational timelines; and
- reviewable incident summaries.

5.5 Utilities and remote infrastructure

Potential workflows include:

- field crew check-ins;
- asset inspection notes;
- dispatch updates;
- outages;
- damage reports;
- work progress;
- location-linked records;
- remote-site history; and
- shift or contractor handoffs.

5.6 Public lands and field services

Potential workflows include:

- ranger communications;
 - trail and road conditions;
 - visitor incidents;
 - maintenance updates;
 - remote team coordination;
 - natural-resource observations;
 - weather-related access changes; and
 - interdepartmental handoffs.
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6. Demonstrations and Validation Environments

TerraSatch uses controlled and illustrative demos to test workflows, mapping, event structure, and operational UX.

A demo associated with a geographic area, agency name, resort name, avalanche center, or use case should **not** be interpreted as an endorsement, procurement commitment, formal deployment, or official partnership unless TerraSatch and the named organization have explicitly announced that relationship.

Public and controlled TerraSatch environments

Main platform

- TerraSatch: <https://www.terasatch.com/>
- TerraSatch data room / controlled materials: <https://data.terasatch.com/>
- TerraSatch API: <https://api.terasatch.com/>
- TerraSatch Edge: <https://terasatch.com/edge>

Avalanche and mountain demonstrations

- AvyTS: <https://avy.terasatch.com/>
- Utah avalanche workflow demonstration: <https://uac.terasatch.com/>
- Colorado avalanche workflow demonstration: <https://caic.terasatch.com/>
- Snowbird mountain-operations demonstration: <https://snowbird.terasatch.com/>

Wildfire demonstration

- PyroTS: <https://pyro.terasatch.com/>

Remote-expedition demonstration

- Everest / high-altitude field-operations demonstration: <https://everest.terasatch.com/>

How these demos should be described

Preferred language:

- controlled demonstration;
- illustrative product preview;
- workflow prototype;
- integration preview;
- field-validation environment; or
- partner-specific concept environment where applicable.



Avoid language such as:

- “official partner”;
- “deployed with”;
- “used by”;
- “endorsed by”; or
- “production system for”

unless that status is documented and publicly approved by the organization involved.

7. Data Integrity, Review, and Governance

7.1 Source integrity

Every meaningful event should retain its source context.

For radio-derived records, that may include the source device, configured channel, site, timestamp, and original approved media or event reference.

For manual observations, it may include the author, time, location, attachments, and organization.

7.2 AI interpretation is a separate layer

AI-generated transcript cleanup, entity extraction, geolocation suggestions, summaries, classifications, and recommended follow-up should remain distinguishable from the source record.

The system should make it possible for a reviewer to see what is:

- original;
- machine-generated;
- human-corrected;
- human-approved; and
- published.

7.3 Human-in-the-loop controls

Organizations should be able to configure review requirements around higher-impact outputs.

Examples include:

- transcript correction;
- observation approval;
- location confirmation;



- high-severity event review;
- map publication;
- briefing approval; and
- alert release.

7.4 Organization and site isolation

Operational records should remain scoped to the organization and site that owns them unless an authorized sharing workflow exists.

Cross-organization access should be explicit and governed by the relevant agreement.

7.5 Retention

Retention should be configurable according to:

- organization policy;
- data type;
- operational need;
- legal requirements;
- privacy requirements;
- media storage cost; and
- contract terms.

Not every raw audio segment needs the same retention policy as a reviewed observation or final incident report.

7.6 Security

Security controls should include, as applicable:

- encrypted transport;
- protected stored data;
- scoped API credentials;
- organization/site/device access controls;
- role-based user access;
- audit history;
- credential rotation;
- controlled sharing; and
- deployment-specific retention.

Specific security certifications should not be claimed until formally achieved.



8. Safety and Operational Boundaries

TerraSatch is decision-support software.

It does not guarantee safety.

It does not replace:

- official avalanche advisories;
- professional avalanche forecasting;
- ski-patrol judgment;
- incident command;
- qualified wildfire personnel;
- SAR command;
- engineering review;
- medical direction;
- land-management authority;
- organizational communications policy; or
- professional training.

8.1 Recommended use principles

Preserve the evidence

Do not collapse the source observation and AI interpretation into one record.

Communicate uncertainty

When location, transcription, classification, or interpretation is uncertain, the system should show that uncertainty.

Review high-impact outputs

Organizations should determine who is authorized to approve alerts, briefings, map changes, and operational reports.

Keep existing authoritative sources visible

Official forecasts, warnings, incident products, and agency data should remain clearly identified as authoritative where applicable.

Record corrections

Corrections should update the working record without erasing the fact that a change occurred.



9. Pilot and Commercial Model

TerraSatch's current go-to-market model begins with a constrained TerraListen discovery period.

9.1 30-day limited pilot

Price: \$0

The purpose of the pilot is to prove one useful workflow without creating the cost or support burden of a premature production deployment.

Current public scope:

- one small team;
- up to five participants;
- one authorized workflow;
- one primary language;
- up to two hours of approved sample radio audio;
- one findings summary;
- no custom hardware;
- no custom integration; and
- no production SLA.

The pilot should answer a practical question:

Does turning this specific field communication into a structured, reviewable operational record save enough time or improve enough continuity to justify a rollout?

9.2 Team rollout

Pricing: scoped quote

A paid team deployment should be based on measured operating requirements rather than invented per-seat precision.

Primary pricing drivers include:

- number of active radios;
- number of channels;
- number of teams;
- number of sites;
- hours of audio processed;
- retention volume;
- approved documents and operational memory;
- language requirements;
- integrations;



- reporting requirements; and
- support coverage.

9.3 Organization deployment

Pricing: annual scoped agreement

Organization-level deployments may include:

- multiple sites;
- multiple departments;
- higher data volume;
- additional languages;
- data-governance requirements;
- custom integrations;
- implementation support;
- retention policy;
- reporting workflows; and
- higher-assurance operational support.

9.4 Hardware

TerraSatch should remain hardware-flexible.

The product can support known-compatible receiver and field-compute configurations without requiring TerraSatch to manufacture every device.

Where hardware is supplied or recommended, the commercial agreement should distinguish:

- TerraSatch software support;
 - configuration and deployment support;
 - third-party manufacturer warranty; and
 - organization responsibility for licensed or authorized radio operation.
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10. Validation and Success Metrics

TerraSatch should measure whether the workflow is useful before making broad claims about safety outcomes.

10.1 TerraListen workflow metrics

Useful metrics include:



- percentage of approved audio that produces a usable event;
- transcript review rate;
- correction rate;
- location-resolution success;
- time from event to review-ready record;
- duplicate/noise suppression;
- report-generation time;
- review latency;
- event-to-map linkage rate;
- handoff completeness; and
- operator acceptance.

10.2 Operational impact metrics

Pilot teams can evaluate:

- time saved writing summaries;
- time saved reconstructing incident timelines;
- reduction in duplicated manual transcription;
- fewer missed handoff items;
- improved traceability from report to source;
- faster retrieval of past observations;
- reduced effort preparing daily or incident documentation; and
- integration effort compared with replacing existing systems.

10.3 Terrain and module validation

Module-specific claims should be validated separately.

Examples include:

- cell geometry accuracy;
- location precision;
- terrain-derivative correctness;
- agreement with authoritative boundaries;
- observation-to-cell association;
- time alignment with weather and forecast data;
- model calibration where probabilistic models are used; and
- reviewer agreement.

TerraSatch should not use safety incidents “prevented” as a marketing metric unless a rigorous and defensible methodology exists.



11. Roadmap

The roadmap follows the same operating loop as the product.

11.1 LISTEN — harden TerraListen

Near-term priorities include:

- reliable long-running ingest;
- stronger signal/noise segmentation;
- multi-channel deployment patterns;
- source and channel separation;
- clearer device/site pairing;
- organization-level credentials and permissions;
- improved edge diagnostics;
- repeatable Windows deployment;
- continued Linux support;
- additional supported edge environments;
- better event review; and
- measured processing and retention cost.

11.2 WATCH — deepen map and terrain context

Priorities include:

- more accurate place-name and terrain resolution;
- improved cell visualization;
- clearer map legends;
- map-first layer controls;
- responsive terrain-aware UX;
- seasonal and incident time views;
- wind and weather context;
- provider geometry preservation;
- local-to-regional context;
- drone-derived cells; and
- GIS-friendly exports.

11.3 LEARN — strengthen operational memory

Priorities include:

- source-linked event history;
- documents connected to incidents and cells;
- snowpit and field-observation continuity;
- shift and season views;



- searchable organizational history;
- clearer reviewer correction history;
- policy-aware Satchy workflows; and
- improved summaries of what changed.

11.4 ADAPT — expand approved outputs and integrations

Priorities include:

- stronger handoff generation;
- briefing templates;
- incident packages;
- reviewed alerts;
- task and follow-up workflows;
- document generation;
- map update workflows;
- organization-specific exports;
- webhooks and integrations; and
- partner API patterns.

11.5 Broader platform expansion

After the TerraListen workflow is proven and repeatable, TerraSatch can extend the same infrastructure into deeper:

- AvyTS workflows;
 - PyroTS workflows;
 - HydroTS workflows;
 - GeoTS workflows;
 - InfraTS workflows;
 - drone-assisted WATCH capabilities;
 - multi-site operational memory; and
 - partner-facing field-intelligence integrations.
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12. Integration Philosophy

TerraSatch should be easiest to adopt when it complements existing systems.

12.1 Radio-first does not mean radio-only

TerraListen starts with radio because radio remains a common source of time-sensitive field information.



The same event architecture can support other approved inputs, including:

- operator forms;
- incident systems;
- sensors;
- GIS events;
- documents;
- drone observations; and
- partner APIs.

12.2 Map-agnostic

TerraSatch should integrate with existing map and GIS environments rather than require a single proprietary map provider.

12.3 API-first

Approved organizations and technical partners should be able to move reviewed TerraSatch records into the tools they already operate.

12.4 Human-authorized

Integrations should preserve organizational control over what is sent, published, retained, or acted on.

13. External Data and Reference Sources

TerraSatch may use public data, organization-provided data, and partner-provided data according to source terms and deployment agreements.

The following are examples of authoritative or commonly used public resources relevant to TerraSatch workflows.

Avalanche operations

- Avalanche.org: <https://avalanche.org/>
- National Avalanche Center: <https://avalanche.org/national-avalanche-center/>
- American Avalanche Association: <https://www.americanavalancheassociation.org/>
- Snow, Weather and Avalanches: Observation Guidelines (SWAG):
<https://www.americanavalancheassociation.org/swag>



Weather

- National Weather Service: <https://www.weather.gov/>
- NWS Active Alerts: <https://www.weather.gov/alerts>

Terrain and elevation

- USGS 3D Elevation Program: <https://www.usgs.gov/3d-elevation-program>
- The National Map / 3DEP tools: <https://apps.nationalmap.gov/3depdem/>
- OpenTopography: <https://opentopography.org/>
- OpenTopography Data: <https://opentopography.org/data>

Hydrology and snow telemetry

- USGS Water Data for the Nation: <https://waterdata.usgs.gov/>
- USGS National Water Dashboard: <https://dashboard.waterdata.usgs.gov/>
- NRCS Snow and Water Interactive Map:
<https://www.nrcs.usda.gov/resources/data-and-reports/snow-and-water-interactive-map>
- NRCS SNOTEL historic data: <https://wcc.sc.egov.usda.gov/nwcc/tabget>

Wildfire

- National Interagency Fire Center: <https://www.nifc.gov/>
- NIFC Incident Information: <https://www.nifc.gov/nicc/incident-information>
- Remote Automatic Weather Stations: <https://raws.nifc.gov/>
- LANDFIRE: <https://www.landfire.gov/data>
- LANDFIRE fuel products: <https://www.landfire.gov/fuel>

Partner and proprietary data

TerraSatch should only identify an organization's private or partner data as an active TerraSatch source when that data has been explicitly provided, authorized, or integrated under an appropriate agreement.

Publicly accessible information from an organization should not be described as a formal data partnership merely because TerraSatch can reference or ingest it.

14. Company and Team

TerraSatch Inc.



TerraSatch is a Utah-built field-intelligence company focused on backcountry teams, mountain operations, remote field work, and other environments where communications and operational context frequently extend beyond reliable network coverage.

The company's product direction is summarized by:

LISTEN. WATCH. LEARN. ADAPT.

Keaton McCune

Founder and Director of Field Intelligence.

Focus areas include TerraListen, TerraSatch Edge and API infrastructure, field-intelligence workflows, geospatial systems, radio-to-record processing, operational UX, and partner validation.

LinkedIn: <https://linkedin.com/in/keaton-m>

Ericka Downs

Co-founder and operations-focused contributor supporting field operations, deployment planning, workflow validation, and company development.

Team descriptions should be updated when roles or formal responsibilities change.

15. What TerraSatch Is — and Is Not

TerraSatch is

- field-intelligence infrastructure;
- an AI-assisted radio-to-record workflow;
- terrain-aware operational context;
- a way to preserve source-linked observations;
- a reviewable operational memory system;
- an integration layer;
- a reporting and handoff workflow;
- a platform designed around human judgment; and
- a broader framework for avalanche, wildfire, hydrology, geology, infrastructure, and remote operations.

TerraSatch is not

- an autonomous incident commander;
- an avalanche forecast authority;



- a replacement for official warnings;
 - a replacement for ski patrol;
 - a replacement for SAR command;
 - a replacement for professional wildfire personnel;
 - an engineering certification product;
 - a guarantee of safety;
 - a covert monitoring platform;
 - a requirement to replace existing radios or GIS tools; or
 - a claim that every TerraSatch module is already deployed at production scale.
-

16. Naming and Publication Standard

All new public TerraSatch material should use the current naming hierarchy.

Company

TerraSatch

Operating loop

LISTEN. WATCH. LEARN. ADAPT.

Current radio product

TerraListen

Current AI agent

Satchy

Field-side software

TerraSatch Edge

Platform API and control plane

TerraSatch API

Terrain/module family



TerraGrid, AvyTS, PyroTS, HydroTS, GeoTS, InfraTS

Historical names

Wasatch Relay may be described as an earlier TerraListen MVP.

SherpAI should be treated as historical branding where necessary. Current public material should use Satchy.

17. Publication Notes

This whitepaper intentionally avoids unsupported scale metrics, fabricated precision, placeholder IDs, mock API schemas, unverified performance claims, and language that could imply a formal partnership where only a conversation, controlled demo, or validation environment exists.

When a capability is still under development, it should be labeled as a roadmap item, controlled demo, illustrative product preview, or pilot capability.

When a capability is active infrastructure, it should be described according to what is actually operating.

When a named organization appears in a demo environment, the document should clearly distinguish the demonstration from any formal commercial or operational relationship.

18. Contact and Current Resources

Website

<https://www.terrasatch.com/>

Data room / controlled materials

<https://data.terrasatch.com/>

API

<https://api.terrasatch.com/>

Edge

<https://terrasatch.com/edge>

AvyTS

<https://avy.terrasatch.com/>

PyroTS



<https://pyro.terasatch.com/>

Avalanche workflow demos

<https://uac.terasatch.com/>

<https://caic.terasatch.com/>

Mountain-operations demo

<https://snowbird.terasatch.com/>

Remote-expedition demo

<https://everest.terasatch.com/>

Disclaimer

TerraSatch provides field-intelligence and decision-support software intended to improve information continuity, terrain awareness, documentation, and operational coordination.

TerraSatch does not prevent accidents and does not replace professional judgment, training, official advisories, emergency procedures, incident command, engineering review, medical direction, or organization-specific safety protocols.

Terrain hazards, weather, snowpack, wildfire behavior, hydrology, communications, and remote field operations involve uncertainty and rapidly changing conditions. Users remain responsible for verifying information against authoritative sources, direct field observations, and the procedures of the organization responsible for the operation.

Product capabilities, integrations, and roadmap items may vary by deployment and contract. Controlled demonstrations and illustrative previews are not representations of formal partnerships or production deployments unless explicitly identified as such by TerraSatch and the organization involved.